

Easy-Moderate

Mathematics Advanced12

Practice Questions
(Paper 1)

Name:_____

1. Rationalise the denominator : $\frac{2+\sqrt{3}}{8-2\sqrt{3}}$ (Show your complete solutions)

2. Differentiate the following with respect to x .

(i) $\frac{1}{3x}$

(ii) $e^x \cos x$

(iii) $\ln(\sin x)$

(iv) $(3x^2 - 5x)^6$

3. Find the limiting sum of $1 - \frac{1}{3} + \frac{1}{9} - \frac{1}{27} + \dots$

4. A function is defined by $f(x) = x^3 - 3x^2 - 9x + 22$.

(i) Find the stationary points and their nature.

5. Find the domain of the function $f(x) = \frac{1}{\sqrt{4x^2 - 1}}$.

6. Simplify $\frac{x^2y - xy^2}{x^2 - y^2}$.

7. Find the equation of the tangent to the curve $f(x) = e^{1-2x}$ at the point where $x = \frac{1}{2}$. Write the equation in general form.

8. Find the equation of the tangent to the curve $y = x^3 - 4x^2$ at the point $(2, -8)$.

9. Evaluate $\int_0^{\frac{\pi}{8}} \sec^2 2x \, dx$.

10. Find $\int 3 + \sin 4\pi x \, dx$

11. Solve $\ln(7x - 12) = 2 \ln x$.

12. Find the equation of the tangent to the curve $y = e^{x+2}$ at the point $A(-2, 1)$.

13. Differentiate $y = \frac{x^2}{1-e^x}$

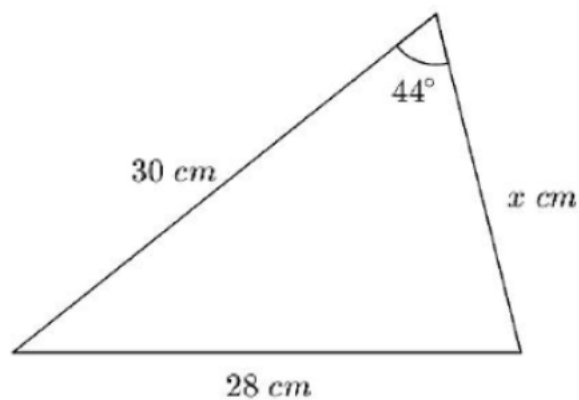
14. Express $\frac{3 + \sqrt{2}}{6 + \sqrt{2}}$ with a rational denominator.

15. Solve $2 \sin \theta - 1 = 0$ for $0 \leq \theta \leq 2\pi$.

16. Find $\int \frac{x}{x^2 - 1} dx$

17. Find $\int \frac{5x^2}{x^3 + 1} dx$

18. The area of the triangle shown is 250 cm^2 .

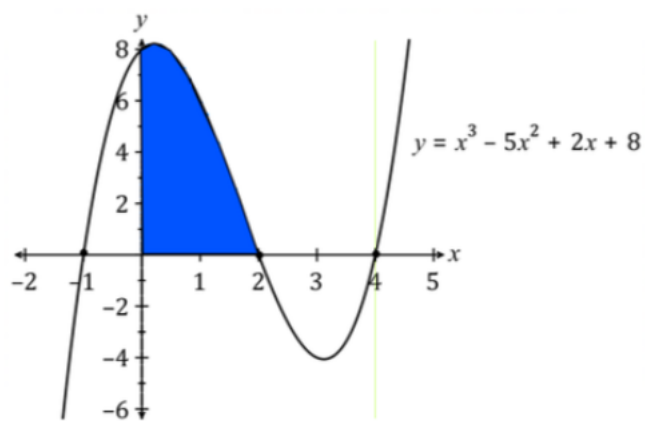


What is the value of x to the nearest whole number?

19. Find $\frac{d}{dx}(x^3 \tan 2x)$

20. Find $\int \frac{2x+2}{4x^2+8x+1} dx$.

21.



Find the shaded area enclosed by the curve $y = x^3 - 5x^2 + 2x + 8$ and the coordinate axes.

22. Find $\int 5x^2(x^3+1)^{10} dx$.